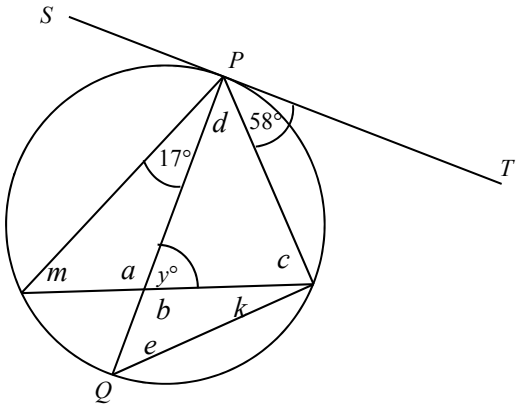
Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	16.5 or 16.49...	**3**	**M2** for $\frac{1.13-0.97}{0.97}[\times 100]$ oe or $\frac{1.13}{0.97}\times 100$ oe or **M1** for $\frac{1.13}{0.97}$ oe
1(b)(i)	35	**2**	**M1** for $60 \div (5+7)$
1(b)(ii)	140	**1**	
1(c)	$1.26 final answer	**3**	**B2** for 1.259... or 1.26 but not as final answer or **M1** for $2.25 \div 0.9416$ If 0 scored, **SC1** for 1.13 × 0.9416
1(d)	15[.0…]	**3**	**M2** for $\sqrt[21]{\frac{58000}{1763000}}$ oe or **M1** for $58000 = 1763000\ (k)^{21}$
1(e)	1239.75	**2**	**B1** for 43 + 0.5 or 28 + 0.5 oe seen
2(a)	103	**3**	**M1** for angle *ABC* or angle *ACB* = $\frac{1}{2}(180-26)$ oe **M1** for angle *ABF* = 26 or angle *CBD* or angle *FBE* = 77 or exterior angle *ACB* = 103 correctly identified or in correct position

Question	Answer	Marks	Partial Marks
2(b)	75	**5**	**B4** for 105 at *a* or *b* **or** 73 at *c* **and** 32 at *d* or **B3** for 58 at *m* **or** 58 at *e* **and** 17 at *k* or **B2** for 32 at *d* **and** 90 soi at (*c*+*k*) **or** 32 at *d* **and** 17 at *k* **or** 73 at *c* or **B1** for 90 soi at (*c* + *k*) or between tangent and radius **or** 32 at *d* **or** 17 at *k* 
3(a)	$1 - r$	**1**	
3(b)(i)	$(1 - r)(1.3 - r)$ [= 0.4]	**1**	**FT** *their***(a)** dep on **(a)** being an expression in *r*
3(b)(ii)	$1.3 - 1.3r - r + r^2$ or better nfww	**M1**	**FT** *their* **(b)(i)**
	$0.9 - 2.3r + r^2 [= 0]$ OR $13 - 13r - 10r + 10r^2 = 4$ oe	**M1**	**Strict FT** *their* expansion to a quadratic then equating to 0.4 and then collecting to 3 terms on 'one side' OR **Strict FT** *their* expansion to a quadratic = 0.4 all multiplied by 10
	$10r^2 - 23r + 9 = 0$	**A1**	no errors or omissions seen